

From Theory to Desk: Quant Tricks That Actually Matter

A Practical Guide to Greeks, Vol Surface/Smiles, YC in IR, Time-Series Models, Numerical Methods & Monte Carlo and Risk/XVA

Amit Kumar Jha, Quant Analyst at UBS

December 11, 2025

Contents

How to Use This Book	2
Greeks as PnL Decomposition Tools	4
Vol Surface, Smile & Arbitrage Tricks	13
Yield Curve & IR Tricks	22
Time-Series & Econometrics Tricks for Quants	29
Numerical Methods & Monte-Carlo Tricks	41
Risk & XVA Tricks	52



5. Why does convexity matter more for long-dated bonds? Convexity scaling. **Answer:** "Convexity scales with maturity squared. 30Y bond convexity 900, 5Y 25. Long bonds have high convexity = valuable in volatile rates environment. If rates are uncertain, you want to be long convexity: you gain more if rates fall than you lose if rates rise. That's why long bonds trade rich in volatile regimes."

6. What happens to a 5Y swap's DV01 as rates rally 100bp? Convexity effect. **Answer:** "DV01 increases as rates fall (positive convexity). If rates rally from 3

7. How would you trade a bull flattening? Fed cuts, long end rallies less. **Answer:** "Bull flattening = rates down, curve flatter (front rally more than back). Trade: long 2Y, short 10Y (flattener). Ratio by DV01: long \$4.5M 2Y, short \$1M 10Y. Also can do receive 2Y swap, pay 10Y swap. Macro: Fed cuts aggressively, market pricing recession → front end rallies hard."

8. Explain why swaps trade at a spread to government bonds. Swap spread. **Answer:** "Swap spread (e.g., 10Y swap rate - 10Y Treasury) reflects: (1) Bank credit risk (LIBOR/SOFR includes bank credit), (2) Liquidity premium (Treasuries more liquid), (3) Supply/demand (pension demand for long bonds compresses spread), (4) Convexity difference (swap has less convexity). Typical 10Y spread = 10-20bp. Widens in crisis."

9. What is the cheapest-to-deliver (CTD) bond in futures? Basis trade. **Answer:** "Treasury futures allow delivery of multiple bonds. CTD = bond with highest implied repo rate (cheapest to buy and deliver). Often the highest duration bond when rates above 6

10. How do you hedge a butterfly trade? 3-legged trade. **Answer:** Butterfly: long 2Y, long 10Y, short $2 \times 5Y$.

Hedge ratios:

DV01 neutrality:

$$DV01_{2Y} + DV01_{10Y} = 2 \times DV01_{5Y}.$$

Key-rate neutrality: hedge the 5Y bucket exposure separately.

Practical rule: choose notionals so that the 2Y and 10Y legs have the same weighted DV01, and the 5Y short is twice as large.

For a \$1M 5Y short, the approximate hedge is:

$$N_{2Y} = 0.44 \text{ M}, \quad N_{10Y} = 0.22 \text{ M}.$$

So you go long \$0.44M 2Y, long \$0.22M 10Y, and short \$1M 5Y.

13. Mini Curve Exercises

Exercise Method For each scenario: compute PnL, identify risk, suggest hedge.

Exercise 1: Short \$2M 5Y bond (DV01 = \$450/M), rates +25bp

- **PnL:** $\$2M \times \$450 \times 25bp = +\$22.5k$ (short bond = profit when rates up)
- **Risk:** Duration risk, if rates reverse you lose.
- **Hedge:** Buy 2Y future: need $\$2M \times (450/200) = \$4.5M$ 2Y to be DV01 flat.

Exercise 2: Long \$5M 10Y (DV01 = \$900/M), steepener view (2Y -30bp, 10Y +20bp)

- **PnL:** $5M \times [900 \times 20bp + 0 \times (-30bp)] = +\$90k$ (long back end wins)
- **Risk:** If flattener instead: lose \$90k.
- **Hedge:** Short \$2.5M 2Y to make it pure steepener bet (DV01 neutral).

Exercise 3: Butterfly: long 2Y \$1M, long 10Y \$1M, short 5Y \$2M. 5Y rate +15bp, others unchanged

- **PnL:** $-\$380/M \times 15bp \times \$2M = -\$11.4k$ (short 5Y loses)

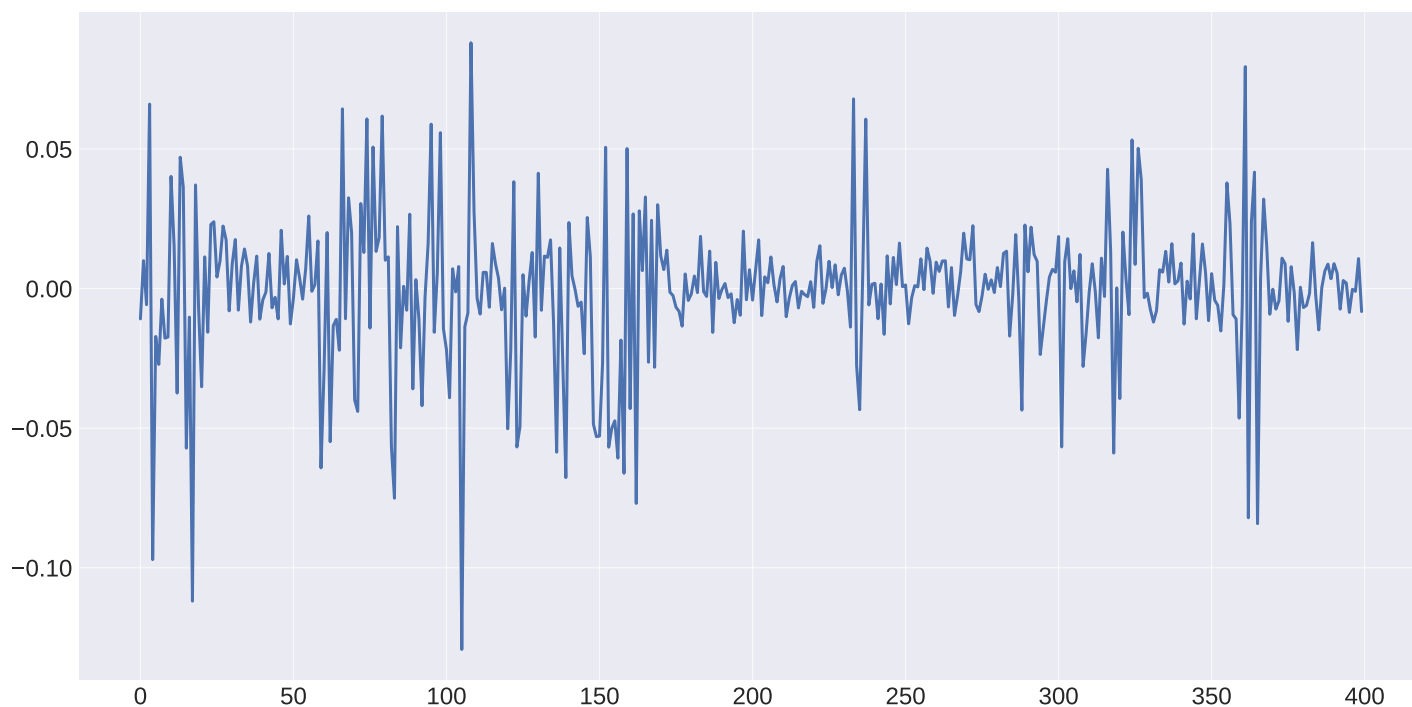


Figure 6: **Regime-Switching Returns**. Green zones = low vol regime (calm). Red zones = high vol regime (crisis). Note volatility clustering and persistence.

11. Simple Factor Models: The Risk Decomposer

Linear Factor Model:

$$r_{i,t} = \alpha_i + \beta_{i,1}F_{1,t} + \beta_{i,2}F_{2,t} + \dots + \epsilon_{i,t}$$

Common Factors:

- **Market:** SPX excess return (CAPM)
- **Style:** SMB (size), HML (value), MOM (momentum)
- **Macro:** Term spread, credit spread, USD strength

"Factor Models are X-Rays for Portfolios" **Memory:** Each factor is a **color**. Your portfolio return = mix of colors.

- **Beta to SPX:** Blue (market risk)
- **Beta to HML:** Red (value tilt)
- **Alpha:** White (unexplained magic)

Use for:

- **Risk decomposition:** "Why did my book lose 2%? 1.5% from market, 0.5% from rates."
- **Hedging:** Hedge factor exposures, not individual assets
- **Performance attribution:** Did I generate alpha or just factor beta?

12. Forecasting vs Backtesting Pitfalls: The Graveyard

Look-Ahead Bias:

Using future info to predict past $\implies$ impossible profits

Example: Using earnings announcement return to predict pre-earnings drift. In live trading, you don't know the announcement return yet!

- **Explicit:** One wrong step (big Δt) = fall (blow up)
- **Implicit:** Always safe, but computationally expensive
- **Crank-Nicolson:** Best of both, but can oscillate if payoff discontinuous

Boundary Conditions: - $S = 0$: Call = 0, Put = Ke^{-rT} - $S \rightarrow \infty$: Call $\rightarrow S - Ke^{-rT}$, Put $\rightarrow 0$ - **Time T:** $V(S, T) = \text{Payoff}(S)$

9. Free Boundary Problems: The American Exercise

Early Exercise Boundary $S^*(t)$:

For $S > S^*(t)$: $V(S, t) = \text{Payoff}(S)$ (exercise)

For $S \leq S^*(t)$: $V(S, t) > \text{Payoff}(S)$ (hold)

"Free Boundary = Where Option Meets Stock" **Memory:** Think of American put. At high S , it's worth more alive (time value). At low S , it's worth dead (exercise). The boundary $S^*(t)$ is where **time value = 0**. On tree, at each node compare **hold value** vs **exercise value**, take max.

Algorithm: 1. Compute continuation value: $e^{-r\Delta t}(pV_u + (1-p)V_d)$ 2. Compare to exercise value: $\max(K - S, 0)$ 3. Value = $\max(\text{continuation}, \text{exercise})$

Smooth Pasting Condition:

$$\left. \frac{\partial V}{\partial S} \right|_{S=S^*(t)} = \left. \frac{\partial \text{Payoff}}{\partial S} \right|_{S=S^*(t)}$$

Ensures: Delta continuous at boundary.

American Option Trick"Dynamic programming: $\max(\text{hold}, \text{exercise})$ "Free boundary = $S^*(t)$ solves $V(S^*, t) = \text{Payoff}(S^*)$ "On tree: backward induction, at each node take $\max(\text{discounted child}, \text{exercise})$. This automatically finds boundary."

Part 4: Model/Method Matching

10. Which Numerical Method for Which Product: The Cheat Sheet

Product	Method	Why	Alternative / Caveat
Vanilla call	BS closed form	Instant	PDE for verification
American put	Binomial tree (100 steps)	Early exercise, fast	PDE if high precision needed
Digital	MC (importance sampling)	Discontinuous payoff	PDE with smoothing
Barrier (single)	PDE (Dirichlet BC)	Fast, accurate hedge ratios	Tree if American
Barrier (double)	MC or PDE (2D)	2 factors	Tree if rates fixed
Asian (arith)	MC (pathwise)	Path-dependent	PDE + aux variable (slow)
Asian (geom)	Closed form	Lognormal average	Use as control variate
Basket (3 assets)	MC (Quasi-MC)	3D grid impractical	PDE if European, low vol
Basket (10 assets)	MC only	Curse of dimensionality	Use Brownian bridge
Bermudan swaption	LSM (Longstaff-Schwartz MC)	Early exercise in MC	Tree (Hull-White) if 1-factor
TARN	MC	Knock-on-knock path dep	PDE impossible
Variance swap	MC (log contract)	Model-independent	Approximation formula

"Method Selection is Like Choosing a Vehicle"

- **Bicycle (BS formula):** Short distance, flat road (vanilla)